5 ETH/RS232 interfaces

5.1 Overview

The communication interface on the configurable control system PNOZmulti 2 is used to

- Download and upload the project,
- Read the diagnostic words,
- Read the error stack.

Base unit PNOZ m B1

The base unit PNOZ m B1 has an Ethernet interface integrated in the device.

Base unit PNOZ m B0

The base unit PNOZ m B0 does not contain an integrated interface. Communication modules with interface can be connected:

- **PNOZ m ES RS232**

  The communication module provides a serial RS232 interface
- **PNOZ m ES ETH**

  The communication module provides 2 Ethernet interfaces

5.2 Communication Module PNOZ m ES RS232

The connection to the RS 232 interface of the communication partner and the communication module PNOZ m ES RS232 is established via a null modem cable.

Transmission rate:
19.2 kBit with

- 8 bits of data
- 1 start bit
- 2 stop bits
- 1 parity bit
- Even parity

5.3 Communication Module PNOZ m ES ETH

5.3.1 Introduction

This chapter describes the special features of communication with the expansion module PNOZ m ES ETH via Ethernet and Modbus/TCP. Access to data from the PNOZmulti 2 via tables and segments is described in Chapter Basics [📖 17].

5.3.2 Overview

The expansion module PNOZ m ES ETH provides an interface to Ethernet. Via this interface, the configurable control system PNOZmulti 2 can be connected via Ethernet to control systems that support Modbus/TCP. Modbus/TCP is designed for fast data exchange at field level. The expansion module PNOZ m ES ETH is a passive Modbus/TCP subscriber

(Server). The basic functions of communication with Ethernet or Modbus/TCP conform to IEEE 802.3. The central controller (Client) reads output information from the Servers and writes input information to the Servers as part of each cycle. In addition to the cyclical payload transmission, the PNOZ m ES ETH also has diagnostic functions.

The connections are established via two RJ45 sockets.

The Ethernet interface is configured in the PNOZmulti Configurator (for description see online help for the PNOZmulti Configurator).

The communication module PNOZ m ES ETH supports Modbus/TCP [72].

The module can manage up to 8 Modbus/TCP connections and up to 4 PG port (Port 9000) connections.

5.3.3 Module features

- Can be configured in the PNOZmulti Configurator
- Network protocols: TCP/IP, Modbus/TCP
- Status indicators for communication and display of errors
- Transmission rate 10 MBit/s (10BaseT) and 100 MBit/s (100BaseTX), full and half duplex

5.3.4 Modbus/TCP

There is no need to configure a connection on the PNOZ m ES ETH. Port 502 is used in accordance with the Modbus/TCP specification.

For details see Modbus/TCP [72]

5.4 Integrated Ethernet interface

5.4.1 Overview

The base unit PNOZ m B1 has an integrated Ethernet interface. Via this interface, the configurable control system PNOZmulti 2 can be connected via Ethernet to control systems that support Modbus/TCP. Modbus/TCP is designed for fast data exchange at field level. The PNOZmulti is a passive Modbus/TCP subscriber (Server).

The basic functions of communication with Ethernet or Modbus/TCP conform to IEEE 802.3. The central controller (Client) reads output information from the Servers and writes input information to the Servers as part of each cycle. In addition to the cyclical payload transmission, the PNOZ m B1 also has diagnostic functions.

The connections are established via two RJ45 sockets.

The Ethernet interface is configured in the PNOZmulti Configurator (for description see online help for the PNOZmulti Configurator).

The module can manage up to 8 Modbus/TCP connections and up to 4 PG port (Port 9000) connections.

5.4.2 Module features

- Can be configured in the PNOZmulti Configurator
- Network protocols: TCP/IP
- Status indicators for communication and display of errors
- Transmission rate 10 MBit/s (10BaseT) and 100 MBit/s (100BaseTX), full and half duplex

5.4.3 Modbus/TCP

No connection needs to be configured. Port 502 is used in accordance with the Modbus/TCP specification.

For details see Modbus/TCP [📖 72]

5.5 Communication procedure

The PNOZmulti 2 is always the connection's Server in the communication; the communication partner (PC, SPS) is the Client. With both communication interfaces, the bytes are sent in the same sequence.



INFORMATION

For communication via Ethernet, the Ethernet interface must be set up in the PNOZmulti Configurator. The procedure is described in the PNOZmulti Configurator's online help.

Each communication is started by sending a request to the PNOZmulti 2. Requests are used to receive data from the PNOZmulti 2 or send data to the PNOZmulti 2:

1. Request:

The user sends a request to the PNOZmulti 2 via the communications partner.

2. Response:

The PNOZmulti 2 sends a response to the communications partner after approx. 20 to 30 ms, confirming that the request has been received without error. Data is sent in accordance with the request.

5.6 Telegram structure

The telegram used for communication consists of:

- a header,
- the payload,
- a footer.

User → PNOZmulti					PNOZmulti → User			
		Byte	Request				Byte	Response
Header		0	0x05		Header		0	0x05
		1	0x15				1	0x15
		2	0x00				2	0x00
		3	Telegram length				3	Telegram length
	(BCC)*	4	Request No.			(BCC)*	4	Confirmation/error
		5	Segment No. HB				5	Segment No. HB
		6	Segment No. LB				6	Segment No. LB
		7	0x00				7	Reserved
Payload (Payload)		7+1	Payload Byte 0		Payload (Payload)		8	Payload Byte 0
		7+2	Payload Byte 1				9	Payload Byte 1
		…	…				…	…
		7+x	Payload Byte x				7+y	Payload Byte y
Footer		7+x+1	BCC		Footer		7+y+1	BCC
		7+x+2	0x10				7+y+2	0x10

x stands for the request's payload amount

y stands for the response's payload amount

*Check sum field

5.6.1 Header

Byte 0 ... Byte 7 form the data block's header

- Byte 0: Always 0x05
- Byte 1: Always 0x15
- Byte 2: Always 0x00
- Byte 3: Telegram length = Length of check sum area +1
- Byte 4
  - Request: Request number
    A request is defined via the request number
    Requests [📖 60]

– Response: Confirmation of request
The request is confirmed: Request number + 0x80 (Bit 7 set).
If the request cannot be processed, an error message is returned
Troubleshooting [📖 70].

- Byte 5: High Byte of segment number
- Byte 6: Low Byte of segment number
- Byte 7
  – Request: Always 0x00
  – Response: Reserved

5.6.2 Payload

The telegram contains payload from Byte 8. The content and number of payload bytes depend on the request. 0 - 40 bytes of payload can be transmitted.

If there is no payload, then the footer directly follows the header, without any payload in between.

- Bytes 8 ... x + 7 (request): Application data, which is sent to the PNOZmulti
- Bytes 8 ... y + 7 (response): Application data, which is sent by the PNOZmulti

5.6.3 Footer

The Byte x + 8 contains a byte-size check sum (Block Check Character = BCC). The value at Byte x + 9 contains the fixed byte value 0x10.

The check sum is calculated as the sum total of the check sum field:

$$BCC = 0 - \left(\sum_{\text{Byte 4}}^{\text{Byte x+7}} \text{byte values}\right)$$

Sample of the check sum field:

$$0 = BCC + \left(\sum_{\text{Byte 4}}^{\text{Byte x+7}} \text{byte values}\right)$$

Example for 2 Bytes payload length

Table 91 Segment 40 is requested:

Telegram: 05 15 00 07 2F 00 00 00 5B 28 [BCC] 10

Calculation of check sum:

1. Total: 2F + 00 + 00 + 00 + 5B + 28 = B2
2. Invert (two's complement): -B2 = 4E
3. Result BCC = 4E

5.7 Description of payload

This section describes the payload that can be transmitted as the result of a corresponding request.

5.7.1 Virtual inputs (Input Byte 0 ... Input Byte 15)

The virtual inputs are defined by the communication partner and transferred to the PNOZmulti 2 . Each input has a number, e.g. input bit 4 of input byte 1 has the number i12.

Input Byte								
0	i7	i6	i5	i4	i3	i2	i1	i0
1	i15	i14	i13	i12	i11	i10	i9	i8
2	i23	i22	i21	i20	i19	i18	i17	i16
…	...	...	...	...	...	...	...	...
15	i127	i126	i125	i124	i123	i122	i121	i120

5.7.2 Watchdog

The watchdog is used to monitor the virtual inputs.

If no virtual inputs are sent by a communications partner within a defined watchdog time (watchdog timeout), the PNOZmulti sets the virtual inputs to "0".

The configuration and functionality of the watchdog varies and so is described in the respective requests.

5.7.3 Virtual outputs (Output Byte 0 ... Output Byte 15)

The virtual outputs are defined in the PNOZmulti Configurator. Each output that is used is given a number there, e.g. o0, o5 .... The state of output o0 is stored in bit 0 of output byte 0; the state of output o5 is stored in bit 5 of output byte 0 etc.

Output Byte								
0	o7	o6	o5	o4	o3	o2	o1	o0
1	o15	o14	o13	o12	o11	o10	o9	o8
2	o23	o22	o21	o20	o19	o18	o17	o16
...	...	...	...	...	...	...	...	...
15	o127	o126	o125	o124	o123	o122	o121	o120

5.7.4 LED status

The statuses of the LEDs are stored in a byte.

Status of LEDs with PNOZ m B0:

The LED status of the base unit PNOZ m B0 can be requested directly as follows

- Bit 0 = 1: LED OFAULT is lit or flashes
- Bit 1 = 1: LED IFAULT is lit or flashes
- Bit 2 = 1: LED FAULT is lit or flashes
- Bit 3 = 1: LED DIAG is lit or flashes
- Bit 4 = 1: LED RUN is lit
- Bit 5-7: Reserved

Status of LEDs with PNOZ m B1:

The LED status of the base unit PNOZ m B1 can be requested directly as follows

- Bit 0 = 1: LED OFAULT is lit or flashes
- Bit 1 = 1: LED IFAULT is lit or flashes
- Bit 2 = 1: LED FAULT is lit or flashes
- Bit 3 = 1: LED DIAG is lit or flashes
- Bit 4 = 1: LED RUN FS is lit
- Bit 5: Reserved
- Bit 6 = 1: LED RUN ST is lit
- Bit 7: Reserved

5.7.5 Tables

Further information can be requested in table form.

A table consists of one or more PNOZmulti 2 segments. Each segment is made up of 13 Bytes.

The communications partner requests the required data using the table number and segment number. The PNOZmulti 2repeats the two numbers and sends the requested data.

The following types are available:

Table 20:	Process data, base unit
Table 21:	Process data, right-hand expansion modules
Table 22:	Process data, left-hand expansion modules
Table 23:	Process data, fieldbus module / communication module
Table 24:	Process data, ST expansion module
Table 70:	Diagnostic words
Table 71:	Enable elements
Table 80:	Project Information
Table 90:	Device data, base unit
Table 91:	Device data, right-hand expansion modules
Table 92:	Device data, left-hand expansion modules
Table 93:	Device data, fieldbus module / communication module
Table 94:	Device data, ST expansion module
Table 110:	Connected devices
Table 201:	Advanced data

The content and the version and device-dependent validity of the tables are described in detail in the chapter Service data [📖 86].

5.8 Requests

A request is defined via the request number and segment number.

The following requests are available:

Request No.	Segment No.	Meaning
0x14	0x01	Sending virtual inputs to the PNOZmulti 2
0x14	0x02	Sending virtual inputs and requesting the LED statuses from the PNOZmulti 2
0x2C	0x01	Requesting virtual input and output data from the PNOZmulti 2
0x2C	0x02	Requesting virtual input and output data and the LED statuses from the PNOZmulti 2 Constraint: PNOZ m B1 from V1.10
0x2F	0x00	Requesting the diagnostic data in table form from the PNOZmulti 2
0x5C	0x00	Sending virtual inputs and requesting virtual output data from the PNOZmulti 2 (suitable for a cyclical exchange)

5.8.1 Mask (Mask Byte 0 ... Mask Byte 15)

The mask is used to determine which of the virtual inputs sent in a Byte are to be set. For example, if only inputs i0 to i5 are to be set in Byte 8, 0x3F must be entered in the mask in Byte 24.

5.8.2 Sending virtual inputs to PNOZmulti 2

Request 0x14 Segment 0x01

The communication partner uses this request to send virtual inputs to the PNOZmulti 2.

The mask (Bytes 24 to 39) is used to determine which of the virtual inputs in a Byte are to be set.

Telegram

Byte	Request		Byte	Response
0	0x05		0	0x05
1	0x15		1	0x15
2	0x00		2	0x00
3	0x25		3	0x05
4	0x14		4	
5	0x00		5	0x00
6	0x01		6	0x01
7	0x00		7	0x00
8	Virtual inputs Input Byte 0: i7 to i0		8	0x6B
...	...		9	0x10
23	Virtual inputs Input Byte 15: i127 to i120			
24	Mask Mask Byte 0: i7 to i0			
...	...			
39	Mask Mask Byte 15: i127 to i120			
40	BCC			
41	0x10			

INFORMATION

If a fieldbus module is configured, virtual inputs cannot be activated via the integrated interface. In this case the request will be rejected by the PNOZmulti 2 with error message 0x63 (request cannot be executed).

5.8.3 Sending virtual inputs and requesting the LED states from PNOZmulti 2

Request 0x14 Segment 0x02

The communication partner uses this request to send virtual inputs to the PNOZmulti 2 in exactly the same way as request 0x14 segment 0x01. It also requests the virtual outputs and the LED status from the PNOZmulti 2 .

The mask (Bytes 24 to 39) is used to determine which of the virtual inputs in a Byte are to be set. For example, if only inputs i0 to i5 are to be set in Byte 8, 0x3F must be entered in the mask in Byte 24.

Telegram

Byte	Request		Byte	Response
0	0x05		0	0x05
1	0x15		1	0x15
2	0x00		2	0x00
3	0x26		3	0x16
4	0x14		4	0x94
5	0x00		5	0x00
6	0x02		6	0x02
7	0x00		7	0x00
8	Virtual inputs Input Byte 0: i7 to i0		8	Virtual outputs Output Byte 0: o7 to o0
...	...		...	...
23	Virtual inputs Input Byte 15: i127 to i120		23	Virtual outputs Output Byte 15: o127 to o120
24	Mask Mask Byte 0: i7 to i0		24	LED status
...	...		25	BCC
39	Mask Mask Byte 15: i127 to i120		26	0x10
40	Control Byte			
41	BCC =0 - (Byte 4 + ... + Byte 40)			
42	0x10			

For details of the LED status see chapter entitled LED status.



INFORMATION

If a fieldbus module is configured, virtual inputs cannot be activated via the communication module.

5.8.3.1 Control Byte (Byte 40)

Bit 0 … 2 of the Control Byte contain a watchdog function.

If no virtual inputs are sent by a communications partner within the defined watchdog time (watchdog timeout), the PNOZmulti 2 sets the virtual inputs to "0".

Control Byte 40:

Reserved	Delayed Re-sponse	Error Message	Reserved	Reserved	W-Timer Bit2	W-Timer Bit1	W-Timer Bit0

- Bit 0 - 2: Watchdog Timeout

Watchdog Timer Bit 2	Watchdog Timer Bit 1	Watchdog Timer Bit 0	Watchdog Timeout
0	0	0	Timer deactivated
0	0	1	100 ms
0	1	0	200 ms
0	1	1	500 ms
1	0	0	1 s
1	0	1	3 s
1	1	0	5 s
1	1	1	10 s

- Bit 3 and 4: Reserved
- Bit 5 Error Message: Error message
  If the bit is set to "1", an error stack entry is generated when the watchdog is triggered.
- Bit 6 Delayed Response: Delayed response
  If the bit is set to "1", the response (send virtual outputs) is sent after a delay of one cycle.
- Bit 7: Reserved



INFORMATION

To test if the watchdog is active, set a virtual input continuously to "1".
If this input becomes "0" after the set watchdog timeout has elapsed, then the watchdog is active.



INFORMATION

The watchdog functions of the requests 0x14 segment 0x02 and 0x5C use the same watchdog timer. In other words, the watchdog timer is reset if either request is called.

5.8.4 Requesting virtual input and output data from PNOZmulti 2

Request 0x2C Segment 0x01

The communication partner uses this request to request the state of the virtual inputs and outputs from the PNOZmulti.

Telegram

Byte	Request		Byte	Response
0	0x05		0	0x05
1	0x15		1	0x15
2	0x00		2	0x00
3	0x05		3	0x25
4	0x2C		4	0xAC
5	0x00		5	0x00
6	0x01		6	0x01
7	0x00		7	0x00
8	0xD3		8	Virtual inputs Input Byte 0: i7 to i0
9	0x10		...	...
			23	Virtual inputs Input Byte 15: i127 to i120
			24	Virtual outputs Output Byte 0: o7 to o0
			...	...
			39	Virtual outputs Output Byte 15: o127 to o120
			40	BCC
			41	0x10

5.8.5 Requesting virtual input and output data and the LED statuses from PNOZmulti 2

Request 0x2C Segment 0x02

The communication partner uses this request to request the status of the LEDs and the virtual inputs and outputs from the PNOZmulti.

Telegram

Byte	Request
0	0x05
1	0x15
2	0x00
3	0x05
4	0x2C
5	0x00
6	0x02
7	0x00
8	0xD2
9	0x10

Byte	Response
0	0x05
1	0x15
2	0x00
3	0x26
4	0xAC
5	0x00
6	0x02
7	0x00
8	Virtual inputs Input Byte 0: i7 to i0
...	...
23	Virtual inputs Input Byte 15: i127 to i120
24	Virtual outputs Output Byte 0: o7 to o0
...	...
39	Virtual outputs Output Byte 15: o127 to o120
40	LED status
41	BCC
42	0x10

5.8.6 Requesting the diagnostic data in table form from PNOZmulti 2

Request 0x2F

The communication partner uses this request to request data from the PNOZmulti in table form.

The content of the tables and segments is described in detail in the Appendix.

Telegram

Byte	Request		Byte	Response
0	0x05		0	0x05
1	0x15		1	0x15
2	0x00		2	0x00
3	0x07		3	0x14
4	0x2F		4	0xAF
5	0x00		5	0x00
6	0x00		6	0x00
7	0x00		7	0x00
8	Table No.		8	Table No.
9	Segment No.		9	Segment No.
10	BCC		10	Byte 0 of Table x, Segment y
11	0x10		...	...
			22	Byte 12 of Table x, Segment y
			23	BCC
			24	0x10

- Byte 8: Table number

  Example: 0x15 for Table 21: Process data, right-hand expansion modules

- Byte 9: Segment number

  Example: 0x00 for Segment 0, in Byte 4 state of outputs o0 ... o7 of the right-hand expansion modules

INFORMATION

If the requested segment is unavailable, the Segment No. is set to 255.
Example:
Request: Table No. 20 Segment No. 45
Response: Table No. 20 Segment No. 255
Byte 10 ... 22 = 0

5.8.7 Sending virtual inputs and requesting the virtual output data from PNOZmulti 2 (see fieldbus communication)

Request 0x5C

The communication partner uses this request to send input data to the PNOZmulti and request output data from the PNOZmulti (see section entitled Basics [17]).

As with fieldbus communication, 32 Bytes each are reserved for input and output data (Byte 8 – 39); they are updated approx. every 15 ms.

Byte	Request		Byte	Response
0	0x05		0	0x05
1	0x15		1	0x15
2	0x00		2	0x00
3	0x26		3	0x26
4	0x5C		4	0xDC
5	Control Byte		5	Control Byte
6	0x00		6	Reserved
7	0x00		7	0x00
8	Input Byte 0		8	Output Byte 0
9	Input Byte 1		9	Output Byte 1
10	Input Byte 2		10	Output Byte 2
...	...		...	...
39	Input Byte 31		39	Output Byte 31
40	0x00		40	Reserved
41	BCC		41	BCC
42	0x10		42	0x10

5.8.7.1 Input data (to the PNOZmulti)

Input Byte	Content
0	i7 to i0
1	i15 to i8
2	i23 to i16
3	i31 to i24
4	i39 to i32
5	i47 to i40
6	i55 to i48
7	i63 to i56
8	i71 to i64
9	i79 to i72
10	i87 to i80

Input Byte	Content
11	i95 to i88
12	i103 to i96
13	i111 to i104
14	i119 to i112
15	i127 to i120
16	0x00
17	Table number
18	Segment number
19	0x00
20	0x00
21	0x00
22	0x00
23	0x00
24	0x00
25	0x00
26	0x00
27	0x00
28	0x00
29	0x00
30	0x00
31	0x00

The virtual inputs are set and a specific table/segment is requested in the input data.



INFORMATION

If a fieldbus module is configured, virtual inputs cannot be activated via the communication module.

5.8.7.2 Output data (from the PNOZmulti)

Output Byte	Content
0	o7 to o0
1	o15 to o8
2	o23 to o16
3	o31 to o24
4	o39 to o32
5	o47 to o40
6	o55 to o48

Output Byte	Content
7	o63 to o56
8	o71 to o64
9	o79 to o72
10	o87 to o80
11	o95 to o88
12	o103 to o96
13	o111 to o104
14	o119 to o112
15	o127 to o120
16	LED status
17	Table number
18	Segment number
19	Payload Byte 0
20	Payload Byte 1
21	Payload Byte 2
22	Payload Byte 3
23	Payload Byte 4
24	Payload Byte 5
25	Payload Byte 6
26	Payload Byte 7
27	Payload Byte 8
28	Payload Byte 9
29	Payload Byte 10
30	Payload Byte 11
31	Payload Byte 12

The content of the tables and segments is described in detail in the section entitled Service data [86].

5.8.7.3 Control Byte (Byte 5)

Bit 0 … 2 of the Control Byte contain a watchdog function.

If no virtual inputs are sent by a communications partner within the defined watchdog time (watchdog timeout), the PNOZmulti sets the virtual inputs to "0".

Control Byte 5:

Read/ Write	Delayed Re- sponse	Error Message	Reserved	Reserved	W-Timer Bit 2	W-Timer Bit 1	W-Timer Bit 0

▸ Bit 0 - 2: Watchdog Timeout

Watchdog Timer Bit 2	Watchdog Timer Bit 1	Watchdog Timer Bit 0	Watchdog Timeout
0	0	0	Timer deactivated
0	0	1	100 ms
0	1	0	200 ms
0	1	1	500 ms
1	0	0	1 s
1	0	1	3 s
1	1	0	5 s
1	1	1	10 s

- Bit 3 and 4: Reserved
- Bit 5 Error Message: Error message
  If the bit is "1", an error stack entry is generated when the watchdog is triggered.
- Bit 6 Delayed Response: Delayed response
  If the bit is "1", the response (send virtual outputs) is sent after a delay of one cycle.
- Bit 7: Read/Write: Read/write access

  If the bit is "1", write protection is active; data cannot be overwritten. With read access, the watchdog timer is not reset, Bit 6 Delayed Response is deactivated.



INFORMATION

The watchdog functions of the requests 0x14 segment 0x02 and 0x5C use the same watchdog timer. In other words, the watchdog timer is reset if either request is called.



INFORMATION

To test if the watchdog is active, set a virtual input continuously to "1".
If this input becomes "0" after the set watchdog timeout has elapsed, then the watchdog is active.

5.9 Troubleshooting

5.9.1 Request format does not meet specifications

If the request format does not meet the specifications, the PNOZmulti 2 sends the following response:

Byte	Response
0	0x05
1	0x02
2	0x00

Byte	Response
3	0x02
4	0x00
5	0x02
6	0x10

5.9.2 Error while executing a request

If an error occurs while executing a request, the PNOZmulti 2 sends the following response:

Byte	Response
0	0x05
1	0x15
2	0x00
3	0x05
4	Error Code
5	0x00
6	0x00
7	0x00
8	BCC
9	0x10

Error Codes (Byte 4):

- 0x62: Request's BCC is incorrect
- 0x63: Request cannot be executed
- 0x64: Request unknown
- 0x67: Table or segment number unavailable
- 0x68: PNOZmulti 2 not ready